Lab 11: Analysis Report

CS24B055

1 Overview and Objectives

This lab exercise focuses on mitigating control hazards within a two-stage pipelined processor architecture (comprising Stage 1: Fetch/Decode/Register Read, and Stage 2: Execute/WriteBack). To prevent stalling the pipeline at every conditional branch, a dynamic branch predictor was integrated into the design. The core of this implementation is a 64-entry Branch History Table (BHT) that utilizes 2-bit saturating counters to forecast whether a branch will be taken or not taken. Because the decode phase resolves the branch target early in Stage 1, a dedicated Branch Target Buffer(BTB) is unnecessary for this architecture.

2 Architectural Modifications

Integrating the dynamic branch predictor required several targeted updates across the processor's pipeline stages:

- **Stage 1 (Fetch & Decode) – Prediction Logic:**
  - A 64-entry BHT was introduced, which is indexed using bits [7 : 2] of the Program Counter (PC).
  - The processor reads the 2-bit counter associated with the current index. If the Most Significant Bit (MSB) is 1, the branch is predicted as "Taken." If the MSB is 0, it is predicted as "Not Taken."
  - Based on this prediction, the next PC is immediately updated: if "Taken," it jumps to the calculated target address ($PC + Imm\_B$); if "Not Taken," it increments normally to $PC + 4$.
- **Stage 2 (Execute & Writeback) – Outcome Resolution:**
  - Additional logic was implemented to evaluate the true condition of the branch and verify it against the prediction made in Stage 1.
- **Misprediction Recovery:**
  - When a branch is mispredicted, a pipeline flush mechanism is triggered. The erroneously fetched instruction currently in Stage 2 is annulled to prevent unintended memory operations or register writebacks.
  - The PC is subsequently redirected to the accurate target address to resume correct execution.
- **Predictor State Updates:**

– The corresponding 2-bit saturating counter in the BHT is updated during this phase. Depending on the actual branch resolution, the counter increments (shifting toward strongly taken) or decrements (shifting toward strongly not taken).

3 Performance Evaluation

To measure the predictor's efficiency, a series of standard test cases were run on both the original baseline processor and the newly modified pipeline. The overall cycle counts are recorded below:

4 Results

The following results were collected for the MNIST program:

4.1 Without MUL instructions

- Total Cycles (without predictor): 800881
- Total Cycles (with predictor): 779056
- Total Instructions: 591857
- Correct Predictions: 41921
- Mispredictions: 16610

4.2 With MUL instructions

- Total Cycles (without predictor): 198767
- Total Cycles (with predictor): 194046
- Total Instructions: 155047
- Correct Predictions: 5191
- Mispredictions: 448

4.3 Performance Improvement

- Cycle Reduction: 21825
- Percentage Improvement: 2.7 %

5 Bonus Analysis 1: BHT Size Variation

We tested different BHT sizes:

- 64 entries - 779056
- 128 entries - 778880
- 256 entries - 778878

Result Analysis:
The recorded data highlights a substantial boost in processing efficiency for the modified architecture. In the standard baseline model, control hazards inherently mandate pipeline flushes or stalls upon encountering taken branches. By incorporating the 2-bit dynamic predictor, the processor successfully anticipates repetitive branch outcomes (such as loop iterations) with a high degree of accuracy. This anticipation drastically cuts down the frequency of misprediction-induced flushes, directly yielding the reduction in total execution cycles observed during testing.

6 Concluding Remarks

Implementing a 2-bit saturating counter successfully alleviated control hazard penalties in the two-stage pipeline design. By introducing a relatively low-cost 64-entry BHT, the processor gained the ability to adjust to program branch behavior dynamically at runtime. The evaluation metrics confirm that this architectural enhancement limits pipeline flushing and cycle overhead, thereby delivering a notable improvement in the processor's overall throughput and computational efficiency.